

MODULE 12 · DOMAIN 4 - DESIGN COST-OPTIMIZED ARCHITECTURES

Network Cost and Cost Management

Data transfer, NAT gateways, and the tools that show you the bill.

DOMAIN 4 of 4 EXAM WEIGHT 20% MODULE 12 of 22

Data **into** AWS is free. Data **out** costs money, and data crossing a boundary inside AWS often does too.

What data transfer actually costs

PATH	CHARGE
Internet into AWS	Free
AWS out to the internet	Charged per GB, the largest transfer cost for most
Between Availability Zones	Charged in both directions
Between Regions	Charged, and more than cross-AZ
Within one AZ, private IPs	Free
Through a NAT gateway	Hourly charge plus a per-GB processing charge
Via a gateway VPC endpoint	Free. No hourly and no per-GB charge
Out via CloudFront	Charged, but lower than direct S3 or EC2 egress

ON THE EXAM

A gateway endpoint for S3 or DynamoDB removes NAT processing charges entirely. It is the single most repeated network cost answer.

The patterns that reduce network cost

- ◆ **Gateway endpoints for S3 and DynamoDB.** Free, and removes that traffic from the NAT gateway entirely.
- ◆ **CloudFront in front of S3 or an ALB.** Cached requests never reach the origin, and egress is cheaper.
- ◆ **Keep chatty traffic within an AZ** where the architecture allows it.
- ◆ **Interface endpoints** are charged, so they save money only when they displace more expensive NAT traffic.
- ◆ **Cross-Region replication** is a real ongoing transfer cost, not just a storage cost.

ON THE EXAM

If private instances talk to S3 heavily through a NAT gateway, the answer is nearly always a gateway endpoint.

The cost management tools

TOOL	WHAT IT IS FOR
AWS Cost Explorer	Visualise and analyse past spend. Filter and group by tag, service, account
AWS Budgets	Set a threshold and alert, or take action, when spend or usage crosses it
Cost and Usage Report	The most detailed line-item data, delivered to S3 for your own analysis
AWS Cost Anomaly Detection	Machine learning alerts on unusual spending patterns
AWS Trusted Advisor	Checks across cost, performance, security, fault tolerance and limits
AWS Compute Optimizer	Right-sizing recommendations from actual utilisation
Cost Allocation Tags	How spend gets attributed to teams, projects or environments

ON THE EXAM

'Alert me when spend exceeds X' is Budgets. 'Show me where the money went' is Cost Explorer. 'Every line item for our own analysis' is the CUR.

An analytics platform runs EC2 instances in private subnets that read roughly forty terabytes per month from S3 in the same Region. All traffic exits through a NAT gateway.

The bill shows a large NAT gateway data processing charge. The team wants to reduce it without exposing the instances to the internet.

What should the solutions architect recommend?

- ☐ A Create a gateway VPC endpoint for S3
- ☐ B Create an interface VPC endpoint for S3
- ☐ C Move the instances to public subnets
- ☐ D Add NAT gateways in each Availability Zone

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Gateway endpoint for S3
S3 traffic bypasses the NAT gateway entirely, and gateway endpoints have no hourly or per-GB charge at all.
- B** Interface endpoint for S3
Also removes the NAT charge but bills hourly and per GB. At forty terabytes a month that is a large avoidable cost.
- C** Public subnets
Removes the NAT charge by exposing the instances, which the question explicitly rules out.
- D** More NAT gateways
Processing is charged per GB, so more gateways add hourly charges without reducing the processing fee.

THE SIGNAL

This pattern returns from module two because it is both the security answer and the cost answer. Gateway endpoints are free.

A company hosts a video training library on S3 behind an Application Load Balancer and EC2 web servers. Ninety percent of the bill is data transfer out to the internet, serving the same three hundred videos to users worldwide.

Users in Australia and South America also report slow playback.

Which change addresses both cost and performance?

- ☐ A Add CloudFront in front of the application
- ☐ B Move the videos to S3 Glacier Instant Retrieval
- ☐ C Add Auto Scaling to the web tier
- ☐ D Replicate the bucket to Regions in Australia and South America

◆ PAUSE AND DECIDE

Which change addresses both cost and performance?

- A

 CloudFront
The same three hundred videos cache extremely well, so most requests never reach the origin, and CloudFront egress is cheaper than direct egress.
- B

 Glacier Instant Retrieval
Reduces storage cost, which is ten percent of the bill, and adds retrieval charges on frequently watched files.
- C

 Auto Scaling
Addresses compute capacity. The stated problems are egress cost and distance-related latency.
- D

 Replicate to two more Regions
Helps latency in those two places, adds storage and replication cost, and does nothing for users elsewhere.

THE SIGNAL

A small set of files served repeatedly to a global audience is the ideal CloudFront case. It fixes egress cost and latency together.

A finance team needs to know how much each of twenty-two product teams spends on AWS each month. All workloads run in a single shared account.

They need the breakdown to appear in monthly reporting and to be able to alert a team when its spend exceeds an agreed figure.

Which combination achieves this? (Choose two.)

- | | |
|---|---|
| ☐ A Apply cost allocation tags to resources and activate them in the billing console | ☐ B Create AWS Budgets filtered by tag, with alerts per team |
| ☐ C Move every team into its own AWS account | ☐ D Enable AWS Cost Anomaly Detection on the account |

◆ PAUSE AND DECIDE

Which combination achieves this? (Choose two.)

- A

 Cost allocation tags
Tags are how spend is attributed within one account, and they must be activated in the billing console before they appear in reports.
- B

 Budgets filtered by tag
Once tags are active, a budget per team with its own threshold and alert delivers exactly the requirement.
- C

 One account per team
A valid long-term strategy and a large migration. The question asks for reporting and alerting, not restructuring.
- D

 Cost Anomaly Detection
Flags unusual patterns, not spend against an agreed figure. It answers 'is something odd' rather than 'has this team exceeded its limit'.

THE SIGNAL

Tags attribute cost. Budgets alert on it. The detail people forget is that tags must be activated in the billing console.

A three-tier application spreads its web, application and database tiers across three Availability Zones for resilience. The application tier makes very frequent small calls to the database tier.

The bill shows a significant cross-Availability Zone data transfer charge, and the architecture team asks whether resilience should be reduced to save money.

What should the solutions architect advise?

- ☐ A Consolidate all tiers into one Availability Zone to eliminate the charge
- ☐ B Keep the multi-AZ design, and reduce the charge by making the calls less chatty or caching results
- ☐ C Move the database into the same subnet as the application tier
- ☐ D Route the internal traffic through a NAT gateway to consolidate charges

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A** Consolidate into one AZ

Removes the charge by removing the resilience. Trading availability for a transfer saving is not an acceptable answer.
- B** Keep multi-AZ, reduce the chattiness

Attacks the cause, which is call volume, while preserving the availability the design exists for. Caching removes calls entirely.
- C** Same subnet as the app tier

A subnet lives in one AZ, so this is option A wearing different clothes. The database loses its zone independence.
- D** NAT gateway for internal traffic

Adds an hourly charge and a per-GB processing fee to traffic that currently pays neither. It increases cost.

THE SIGNAL

Never trade away a stated resilience requirement for a cost saving. Reduce the traffic instead of removing the redundancy.

A company's monthly AWS bill jumped by eleven thousand dollars with no known change. Leadership wants to know exactly which service, account and resource caused it, and wants to be told automatically if it happens again.

They also want detailed line-item data so their own analysts can investigate independently.

Which combination meets these requirements? (Choose two.)

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|--|--|
| ☐ A Use Cost Explorer to identify the increase, and enable Cost Anomaly Detection for future alerts | ☐ B Enable the Cost and Usage Report delivered to S3 and query it with Athena |
| ☐ C Set an AWS Budget at the previous month's total | ☐ D Enable AWS Config to record resource changes |

◆ PAUSE AND DECIDE

Which combination meets these requirements? (Choose two.)

- A** Cost Explorer plus Anomaly Detection
Cost Explorer finds where the increase came from, and Anomaly Detection alerts automatically on future unusual spend.
- B** CUR to S3, queried with Athena
The Cost and Usage Report is the most granular billing data available, which is what independent analyst investigation requires.
- C** Budget at last month's total
Alerts on crossing a fixed figure, which catches gradual growth badly and says nothing about which resource caused it.
- D** AWS Config
Records configuration changes, not cost. It might show what changed but attributes no spending to anything.

THE SIGNAL

Cost Explorer for investigation, Anomaly Detection for automatic alerting, CUR for raw line-item analysis. Three tools, three distinct jobs.

A company running two hundred workloads across eighteen accounts wants a regular review that identifies idle load balancers, underutilised instances, unassociated Elastic IPs, and service limits being approached.

They want this without building custom tooling, and they have Business Support.

Which service should be used?

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|---|--|
| ☐ A AWS Trusted Advisor | ☐ B AWS Cost Explorer |
| ☐ C Amazon CloudWatch dashboards | ☐ D AWS Well-Architected Tool |

◆ PAUSE AND DECIDE

Which service should be used?

- | | |
|---|---|
| <p>A Trusted Advisor</p> <p>Its checks cover exactly this list: idle load balancers, low-utilisation instances, unassociated Elastic IPs and service limits, with no tooling to build.</p> | <p>B Cost Explorer</p> <p><i>Shows what was spent by service and account. It does not identify specific idle resources or service limits.</i></p> |
| <p>C CloudWatch dashboards</p> <p><i>Displays metrics you configure. Building the checks is exactly the custom tooling the question rules out.</i></p> | <p>D Well-Architected Tool</p> <p><i>A structured review framework you complete by answering questions. It does not inspect resources automatically.</i></p> |

THE SIGNAL

The list of check types is the giveaway. Idle resources plus service limits together means Trusted Advisor, and the full checks need Business or Enterprise Support.

Module 12 takeaways

- ◆ **In is free, out is charged**, and crossing an AZ, Region or NAT gateway usually costs something.
- ◆ **Gateway endpoints are free**. Interface endpoints are not, so they only save when they displace NAT charges.
- ◆ **Never trade a resilience requirement** for a data transfer saving.
- ◆ **Cost Explorer investigates. Budgets alert on a threshold. Anomaly Detection alerts on the unusual. CUR is the raw data.**
- ◆ **Idle resources plus service limits** means Trusted Advisor.

ON THE EXAM

That completes all four domains. The next five modules cover skills the exam tests without naming them.